0Tribhuvan University

Faculty of Humanities and Social Sciences

Local Farmers Marketplace

A PROJECT REPORT

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Department of Computer Application

Kathmandu Business Campus

Bypass, Balaju-16, Kathmandu

In partial fulfillment of the requirements for the Bachelors in Computer Application

Submitted by

Rabin Shrestha
BCA 6th Semester
Reg. No: 6-2-1219-19-2022
Roll No.: 19

Palistha Maharjan
BCA 6th Semester
Reg. No: 6-2-1219-14-2022
Roll No.: 14

Under the Supervision of

Nixon Dangol

Tribhuvan University

Faculty of Humanities and Social Sciences

Kathmandu Business Campus

Bypass, Balaju-16, Kathmandu

Bachelor in Computer Applications (BCA)

SUPERVISOR'S RECOMMENDATION

I hereby recommend that this project is prepared under my supervision by Rabin Shrestha and Palistha Maharjan entitled "**Local Farmers Marketplace"** in the Partial Fulfillment of requirement for the degree of Bachelor in Computer Application is recommended for that final evaluation.

Nixon Dangol
Project Supervisor
BCA Department
Kathmandu Business Campus

Faculty of Humanities and Social Sciences

Kathmandu Business Campus

Bypass, Balaju-16, Kathmandu

Bachelor in Computer Applications (BCA)

LETTER OF APPROVAL

This is certify that this project is prepared by Rabin Shrestha and Palistha Maharjan entitled "Local Farmers Marketplace " in the Partial Fulfillment of requirement for the degree of Bachelor in Computer Application has been evaluated. In our opinion it is satisfactory in the scope and quality as a project for the required degree.

Nixon Dangol

Supervisor

BCA Department

Nixon Dangol

Program Coordinator

BCA Program

External Examiner

Internal Examiner

ACKNOWLEDGEMENT

Every successful project is the result of continuous guidance, encouragement, and support from many individuals who contribute to achieving the final goal. We feel fortunate to have received proper guidance and assistance during the completion of our project.

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We would like to express our special thanks of gratitude to our Program Coordinator, Mr. Nixon Dangol, for his support and help in our personal development and mainly for the successful completion of this project.

We are highly indebted to Kathmandu Business Campus for their guidance and constant supervision as well as for providing the necessary information regarding the project and support in its completion. Our thanks and appreciation also go to our colleagues for their encouragement and support.

Your Sincerely,

Rabin Shrestha

Palistha Maharjan

ABSTRACT

With the rapid growth of technology, many traditional and manual systems are gradually being replaced by automated online platforms. In this context, the Local Farmers Marketplace is developed as a web-based application that provides an easy, fast, and efficient platform for connecting local farmers directly with consumers. Nepal is a developing country, but the number of internet users is increasing every year, and through the use of the internet people can conveniently access fresh agricultural products without the need to visit physical markets. The main objective of this application is to create an interactive and user-friendly platform where farmers can list their products and customers can easily browse, search, and purchase them. This system helps save time and effort for both farmers and buyers by allowing them to connect through an online environment. Users can view different agricultural products, compare options, and select items according to their needs from anywhere. The system also includes a search feature that allows users to quickly find specific products, where the search engine filters available items based on the user's input to make the process simple and convenient. By providing a digital marketplace, the platform aims to support local farmers, increase product visibility, and make fresh produce more accessible to consumers. This system is developed using HTML, CSS, PHP, JavaScript and MySQL technologies.

Keywords**:** Local Farmers Marketplace, Agriculture, PHP, JavaScript, HTML, MySQL.

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LIST OF ABBREVIATIONS

CRUD	Create, Read, Update and Delete
CSS	Cascading Style Sheet
DFD	Data Flow Diagram
ERD	Entity Relationship Diagram
HTML	Hyper Text Markup Language
JS	Java Script
MySQL	My Structured Query Language
PHP	Hypertext Preprocessor
UI	User Interface

Chapter 1: Introduction

1.1 Introduction

There has been an increasing demand for fresh agricultural products in both urban and rural areas of Nepal. People are becoming more aware of healthy food and prefer to buy fresh vegetables, fruits, and other farm products directly from reliable sources. However, most agricultural products are still sold through traditional markets, which often involve middlemen and limited access for farmers to reach a wider number of consumers. Although some online systems exist for selling agricultural products, many of them do not fully meet the needs of local users and often lack a simple and user-friendly interface. Therefore, to create a convenient and efficient platform for buying and selling farm products, we are introducing the Local Farmers Marketplace. This is a web-based application where users can find a variety of agricultural products such as vegetables, fruits, grains, dairy items, and other farm produce directly from local farmers. Traditionally, people purchase these items from nearby markets, but with the help of the Local Farmers Marketplace, users can easily browse and buy products digitally from their homes. The system helps in planning, organizing, and managing agricultural product sales efficiently. It includes a database that stores information about users, farmers, and products, making it easier to record, manage, and retrieve data. The system also manages product listings, orders, purchases, sales, and payments. Through the admin panel, farmers or vendors can manage their products, update stock, and maintain inventory records.

The system allows users to select products based on their needs, and the selected items are added to a shopping cart. Based on the items in the cart, the system automatically calculates the total cost of the purchase. Users can place orders and choose convenient payment methods such as cash on delivery. Customers can also track their orders to ensure they are available at the delivery address when the product arrives. It is expected that this system will fulfill the needs of customers by providing easy access to fresh farm products while also supporting local farmers by giving them a direct platform to sell their produce. Through the admin panel, vendors can add, update, or remove products, as well as manage categories and subcategories of farm products. They can also monitor and manage customer orders effectively.

1.2 Problem Statement

Previously, the buying and selling of agricultural products were mostly done through traditional and manual methods. Farmers usually sold their products in local markets or through middlemen, and most of the activities were handled without any organized digital system. Because of this manual process, several problems arise in the management and distribution of farm products. The major problems of the traditional agricultural marketplace are as follows:

• Farmers often cannot clearly track which products are available or out of stock.
• The process of buying and selling products is time-consuming and inefficient.
• Customers need to physically visit markets to buy agricultural products.
• Local markets may become crowded, especially during peak hours or seasons.
• Different agricultural products may not be available in one place, making it inconvenient for customers.

These problems need to be addressed because they affect the efficiency of product distribution, reduce the effectiveness of farmers' sales processes, and create inconvenience for customers. Therefore, an online system like the Local Farmers Marketplace is necessary to simplify the buying and selling process, improve accessibility, and enhance the overall efficiency of agricultural product management.

1.3 Objectives

The main objectives of this project are mentioned below:

• To develop an online platform connecting local farmers and consumers.
• To allow users to easily view and search for agricultural products.
• To maintain records of farm products and manage stock.

1.4 Scope and Limitations

Scope

The scope of the Local Farmers Marketplace involves both users and the admin. The admin can add, delete, and update farm products and manage stock through the admin panel. Admins can also add and delete product categories and manage orders placed by users. In the context of Nepal, where many fresh farm products are available only in local markets, this platform will help users find and buy agricultural products online. The system facilitates easy management and purchase of farm products. It also includes user registration, which keeps a record of all users. Users can edit their personal information and product requirements, while the admin can manage user accounts and interactions.

Limitations

The system currently has several limitations that affect its overall functionality and user experience. At present, it only supports payment on delivery in cash, which restricts users who prefer digital or online payment methods. Additionally, because the system is web-based, it requires regular updates and maintenance to ensure smooth performance and security. However, users do not have the ability to perform these updates themselves, making them dependent on the developer or administrator. Due to limited development time, only the core and essential features have been implemented in this version of the system, leaving out many advanced functionalities. This may limit the system's efficiency and flexibility for different types of users. Furthermore, the absence of automated update features can lead to outdated system performance over time. The system may also face scalability issues if the number of users increases significantly. Another limitation is the lack of integration with external services such as online payment gateways. Overall, while the system performs its basic functions, there is significant room for improvement in future versions.

1.5 Development Methodology

The development of the Local Farmers Marketplace system follows a structured methodology to ensure the system is efficient, reliable, and meets user requirements. A systematic approach is applied throughout the development process, covering planning, analysis, design, implementation, and testing phases.

The first phase involves requirement gathering and analysis, where the problems faced by users in traditional agricultural marketplaces are identified. Both functional and non-functional requirements are defined to ensure the system meets user needs and performs effectively.

The next phase is system design, where the overall structure of the system is planned. This includes creating diagrams such as Entity-Relationship Diagrams (ERD) and Data Flow Diagrams (DFD) to clearly represent data relationships and system processes. The design phase ensures a clear blueprint for development.

In the implementation phase, the system is developed using appropriate tools and technologies. The features are built according to the defined requirements, ensuring proper integration between different components of the system.

Finally, the system undergoes testing and evaluation. Different testing methods, such as unit testing and system testing, are conducted to identify and fix errors. This ensures the system is reliable, efficient, and ready for deployment.

This structured development methodology helps in building a well-organized and effective Local Farmers Marketplace system.

Chapter 2: Background Study

2.1 Background Study

Today, interest in using proper systems and software is growing rapidly, making it one of the most active research areas in software development. Users increasingly prefer systems that make their daily tasks easier and more efficient. Agricultural products, especially fresh farm produce, are essential for health and daily consumption, and a proper system can help manage and distribute these resources efficiently.

The Local Farmers Marketplace is developed to manage farm products, agricultural items, and produce from local farmers. This system helps vendors manage stock, track orders, and perform operational tasks more efficiently. It provides features such as customer registration and login, admin login, order placement, billing, and inventory management. The system benefits both consumers and farmers by making it easier to buy and sell farm products online, whether for home, business, or community use.

In the near future, purchasing local farm products is expected to be heavily influenced by digital platforms, making this system an essential tool for both ordering products and managing stock. On the Local Farmers Marketplace, the system handles inputs such as customer registration, login credentials, and order details. Operational procedures include validating logins, viewing orders, generating bills, and delivering products.

The system also includes modules accessible only to the Administrator. These modules are designed for managing and monitoring the website, users, orders, and products. One key module is the Product Inventory and Monitoring, where the admin can view products that are in stock or out of stock. The Maintenance module allows the admin to add, update, or delete products and categories. To add new items, the admin inputs required details such as product name, price, delivery charges, and a short description via the admin panel.

In addition to these functionalities, the system is designed to be simple and user-friendly for both farmers and customers. It allows easy navigation for browsing products, placing orders, and managing accounts without technical difficulty. Basic security features such as login authentication are included to protect user information. The system also ensures controlled access to administrative functions for better management.

2.2 Literature Review

Increasing environmental awareness, population growth, and government policies in Nepal emphasize the need for sustainable agricultural practices and efficient use of resources. Small-scale farms and local producers often operate on limited land areas, making proper planning, management, and resource allocation essential. Many farmers in Nepal rely on traditional methods of production and local markets, which limits their access to urban consumers and reduces profit potential. Digital platforms can help bridge this gap by providing farmers with tools for managing farm products, tracking orders, and reaching consumers directly, while also helping consumers access fresh, high-quality agricultural products conveniently.

Research from various sources indicates that a large number of consumers want to buy farm products directly from producers, but they often lack detailed information about product quality, availability, and pricing. Similarly, many farmers face challenges in marketing their produce effectively, managing inventory, and setting competitive prices. Traditional local markets rarely allow consumers to compare prices between different sellers, and transactions are mostly cash-based, creating inconvenience for both buyers and farmers. This makes it difficult to ensure transparency and fairness in trade. **A** Local Farmers Marketplace addresses these challenges by providing a digital platform where customers can view detailed product information, compare prices from multiple farmers, and make secure online payments. By providing these features, the platform helps build trust between buyers and sellers and encourages more direct transactions without intermediaries.

Customer service and satisfaction are critical to the success of such platforms. Ensuring that every customer has a smooth and enjoyable shopping experience helps retain buyers, generate repeat purchases, and encourage word-of-mouth referrals. Support services provide technical guidance, information on product usage, and assistance with order-related issues, which are essential in a country like Nepal, where many consumers and farmers are still adapting to digital platforms. Expanding the variety of farm products available on the platform, providing seasonal updates, and using creative digital marketing strategies are also important for improving engagement and trust among users.

With the advancement of technology in Nepal, online shopping has become increasingly common, especially in urban areas. Consumers prefer digital solutions because they save time, reduce effort, and allow access to a wider range of products. A Local Farmers

Marketplace allows customers to browse and purchase fresh agricultural products from multiple farmers without leaving their homes. Orders are delivered directly to the customer's doorstep, which is particularly useful for individuals with limited time or mobility. Additionally, the platform may include educational resources for consumers and farmers, such as guidelines for storing fresh produce, instructions for handling perishable items, and tips for sustainable farming practices.

International research and examples support the potential of online agricultural marketplaces. Platforms like Farmigo in the United States, The Food Assembly in Europe, and eNAM (Electronic National Agricultural Market) in India demonstrate that digital farmers' markets can increase farmers' income, reduce dependency on intermediaries, and improve transparency for consumers. These platforms provide features such as product catalogues, price comparisons, online ordering, and scheduled delivery or pickup, which significantly improve efficiency in the farm-to-consumer supply chain. While these platforms operate in countries with advanced technological infrastructure, they provide valuable lessons for implementing similar systems in Nepal.

In the Nepalese context, several local initiatives such as Hamro Krishi**,** Mero Krishi, and small-scale e-commerce platforms have begun bridging the gap between rural farmers and urban consumers. These platforms enable farmers to list products online, manage orders, and receive payments digitally. However, adoption is limited by factors such as low digital literacy among farmers, inconsistent internet connectivity in rural areas, and reliance on cash-based payments. Despite these challenges, platforms like these have shown that digital solutions can increase access to markets, improve price transparency, and reduce post-harvest losses by connecting farmers with buyers in real time.

Furthermore, the COVID-19 pandemic highlighted the importance of online agricultural marketplaces. Mobility restrictions forced consumers and farmers to rely on digital platforms for trading, demonstrating the practicality and necessity of such systems. By providing features like secure payments, product comparisons, real-time inventory updates, and delivery tracking, a Local Farmers Marketplace can increase trust in digital transactions and make it easier for farmers to expand their reach to new customers.

Most people in Nepal do not have enough knowledge about the variety of agricultural products, their nutritional value, or proper storage methods. A Local Farmers Marketplace addresses this gap by offering detailed product descriptions, comparisons, and guidance for

both consumers and farmers. Customer support provides technical advice when needed, helping users make informed decisions. By integrating these features, the platform aims to improve the business experience for both farmers and consumers, ensure sustainable trade, and promote access to high-quality agricultural products across the country.

In conclusion, both global and Nepalese experiences indicate that Local Farmers Marketplaces have the potential to transform the agricultural sector by providing direct access to consumers, improving market transparency, and promoting sustainable practices. The success of such platforms depends on digital accessibility, user-friendly design, effective customer support, and integration of secure payment and delivery systems. For Nepal, this represents a significant opportunity to modernize the agricultural market while empowering rural farmers and providing consumers with convenient access to fresh, quality farm products.

Chapter 3: System Analysis and Design

3.1 System Analysis

System Analysis involves understanding the requirements of the system, identifying its objectives, and studying how the system will meet the needs of users and administrators. For a Local Farmers Marketplace, system analysis focuses on both functional and non-functional requirements, feasibility studies, and system modeling using diagrams to ensure a clear understanding before design and implementation.

3.1.1 Requirement Analysis

a. Functional Requirements

Functional requirements define **what the system must do** to meet the needs of users (buyers) and admins. These are essential features that ensure the marketplace works as intended. For the Local Farmers Marketplace, the key functional requirements are:

1. User Registration and Login

- **Purpose:** To allow buyers to create personal accounts to access the marketplace, place orders, and track purchases.
- **Details:**
  - Users register using an email address, phone number, and password.
  - Login ensures secure access to user-specific features such as profile information.
  - Password recovery and verification mechanisms enhance security and convenience.

2. Admin Login

- **Purpose:** To provide a secure interface for administrators to manage the system and its content.
- **Details:**

Admin login is separate from user login and requires secure authentication.

- Only authorized personnel can access management functions such as product, category, and user management.
- Admin sessions may include activity tracking for accountability.

3. Product Management

- **Purpose:** To maintain a complete and updated catalog of farm products for users to browse and purchase.
- **Details:**
  - Admin can add new products with details including name, category, price, quantity, and description.
  - Admin can update product information, such as price changes or quantity updates.
  - Admin can remove products that are no longer available to prevent user confusion.

4. Category Management

- **Purpose:** To organize products into categories for easier navigation and searching.
- **Details:**
  - Admin can add new product (e.g., vegetables, fruits, dairy).
  - Products can be updated or deleted to reflect changes in the marketplace inventory.
  - Helps users filter products quickly based on improving usability.

5. Order Management

- **Purpose:** To handle all aspects of user orders, ensuring smooth transactions from selection to delivery.

- **Details:**

  - Users can place orders by adding products to a shopping cart and confirming the order.

  - Users can view order history and track the status of ongoing orders (e.g., pending, shipped, delivered).

  - Admin can manage all orders, update order statuses, handle cancellations or returns, and generate sales reports.

6. Payment Management

- **Purpose:** To facilitate secure and convenient payment options for users.

- **Details:**

  - Cash on delivery (COD) is supported for all orders, which is suitable for users in Nepal who prefer cash payments.

  - Online payment integration can be added in the future for credit/debit cards, mobile wallets, or e-banking.

  - Payment confirmation and receipt generation ensure transparency and trust in transactions.

7. Inventory Monitoring

- **Purpose:** To maintain accurate stock levels and prevent overselling or running out of products.

- **Details:**
  - Admin can monitor available quantities for each product.
  - Ensures users see only products that are available, improving reliability.

8. Search and Filter Products

- **Purpose:** To help users quickly find the products they need among many listings.
- **Details:**
  - Users can search by product name, or keyword.
  - Filters can include price range, availability, or product type.
  - Improves user experience and reduces time spent navigating the marketplace.

9. User Profile Management

- **Purpose:** To allow users to manage their personal information and delivery details.
- **Details:**
  - Users can update contact information, delivery addresses, and password.
  - Ensures orders are delivered correctly and that users can maintain up-to-date information.
  - Supports personalized recommendations and better communication from the platform.

USECASE DIAGRAM

In Local Farmers Marketplace, there are two actors i.e. user and admin. User can register, login, select products, checkout products and review products. Whereas, admin can manage categories, products, sales and users.

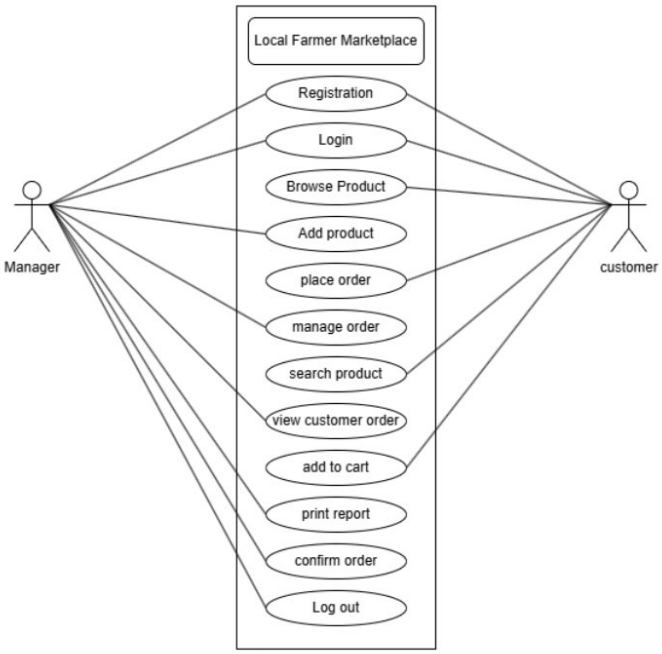

Figure 1 USE CASE DIAGRAM

b.Non-Functional Requirements

Non-functional requirements (NFRs) define how the system performs rather than what it does. They set the standards for usability, reliability, performance, security, scalability, and maintainability, ensuring the system is practical, efficient, and user-friendly. For a Local Farmers Marketplace, these requirements are crucial because the platform serves both farmers and consumers who may have limited technical expertise and expect a smooth, trustworthy experience.

1. Usability

- **Definition:** Usability refers to how easy and intuitive the system is for end-users.
- **Importance:** Farmers and consumers may have varying levels of digital literacy. A system that is simple and easy to navigate encourages adoption, reduces errors, and improves the overall user experience.

- **Implementation:**
  - Clear menus and navigation structure
  - Search and filter options for products

2. Reliability

- **Definition:** Reliability ensures that the system operates correctly under normal conditions and maintains data integrity.
- **Importance:** Users rely on accurate information about product availability, prices, and order status. Any failure or incorrect data can lead to loss of trust, failed transactions, or financial losses.
- **Implementation:**
  - Regular database backups to prevent data loss
  - Error handling for failed orders or system crashes

3. Performance

- **Definition:** Performance measures how quickly and efficiently the system responds to user actions.
- **Importance:** A marketplace must handle multiple users browsing, ordering, and updating inventory simultaneously. Slow performance can frustrate users and lead to abandoned orders.
- **Implementation:**
  - Optimize database queries to fetch product information quickly
  - Use caching for frequently accessed data, such as product lists
  - Monitor system load and response times, especially during peak hours

4. Security

- **Definition:** Security ensures that sensitive information, such as user data, product details, and transaction records, is protected against unauthorized access or attacks.

- **Importance:** Security builds user trust. In a marketplace, users must feel confident that their personal information and payment details are safe. Admin access must also be protected to prevent unauthorized system changes.

- **Implementation:**

  - Password encryption for all user and admin accounts

  - Secure login protocols (e.g., SSL/HTTPS)

  - Regular monitoring for suspicious activity

5. Scalability

- **Definition:** Scalability is the system's ability to grow and handle increased demand without performance degradation.

- **Importance:** As more farmers and consumers join the marketplace, the system must continue to operate efficiently. Planning for scalability ensures long-term sustainability.

- **Implementation:**

  - Design the database and server architecture to support growth

  - Modular system design to easily add new features, such as online payment or delivery tracking

  - Cloud hosting or server expansion to handle more users and transactions

6. Maintainability

- **Definition:** Maintainability is the ease with which the system can be updated, fixed, or improved after deployment.
- **Importance:** Agriculture trends, product types, or user requirements may change over time. The system must be adaptable to support these changes without extensive redevelopment.
- **Implementation:**
  - Use clean, well-documented code for easy updates
  - Modular design for adding or modifying features
  - Monitoring and logging tools to quickly detect and fix issues

3.1.2 Feasibility Analysis

A feasibility study assesses the operational, technical, and economic viability of the proposed project. It helps determine whether the project is practical and worth pursuing before proceeding to the detailed analysis and design phases. From a systems analyst's perspective, the feasibility study is a key tool to recommend whether to continue or discontinue the project. The objective of this feasibility study is to determine if the **Local Farmers Marketplace** can be developed successfully and to suggest alternative solutions if necessary.

i. Technical Feasibility

The Local Farmers Marketplace is technically feasible because the required technologies and hardware are widely available and easy to implement. The proposed system is designed to run on machines that support Windows or any web-enabled platform, making it accessible to users across Nepal. The system uses reliable and proven technologies to ensure smooth performance and scalability.

Main tools and technologies for this system include:

- **Technologies:** HTML, CSS, JavaScript, PHP, MySQL
- **Hardware Requirements:** Laptop or desktop computers; optionally mobile devices for responsive access
- **Software Requirements:** XAMPP/WAMP server, Sublime Text or VS Code, MS Office

The system is technically feasible because these tools are widely available, affordable, and easy to maintain.

ii. Operational Feasibility

The system is user-friendly and can be easily operated by farmers, customers, and administrators with basic computer and internet knowledge. No extensive technical training is required.

- **For Users (Consumers):** The interface is intuitive, allowing them to browse products, place orders, track deliveries, and provide feedback.
- **For Managers:** Managers can manage product listings, monitor , and receive order notifications without difficulty.
- **For Admin:** Administrators can manage users, products, categories, and orders efficiently using the admin panel.

The operational feasibility is high because the system's design prioritizes usability, making it practical for everyday use in the local context of Nepal.

iii. Economic Feasibility

The system is economical to develop and maintain. Development costs are mainly **one-time costs** and include:

- **Personnel Costs:** Developers, testers, and administrators involved in building and maintaining the system.

- **Hardware Costs:** Laptops, servers, and networking equipment.

- **Software Costs:** Development software, server setup, and other tools.

The benefits of the system—connecting farmers directly to consumers, reducing intermediaries, improving price transparency, and increasing sales opportunities—far outweigh the development costs, making the project economically feasible.

iv. Schedule Feasibility

The development and deployment of the Local Farmers Marketplace can be completed within the scheduled timeline. Project planning is done using a Gantt chart, which visually represents tasks against time and ensures that each phase—analysis, design, implementation, and testing—is completed on schedule.

- Proper task allocation ensures timely delivery.

- Regular monitoring prevents delays and keeps the project on track.

Description	Start Date	Duration	End Date	1	2	3	4	5	6	7	8	9	Status
Planning	27-Oct	5	1-Nov										Completed
Analysis	2-Nov	5	7-Nov										Completed
Design	8-Nov	13	22-Nov										Completed
Coding	23-Nov	40	4-Jan										Completed
Testing	5-Jan	7	13-Jan										Completed
Documentation	27-Oct	70	13-Jan										Completed

Figure 2 Gantt Chart of Local Framer Marketplace

3.1.3 Data Modeling

To organize and manage data efficiently, an Entity-Relationship Diagram (ERD) is created for the Local Farmers Marketplace. The system contains the following entities:

1. **Cart** – Contains cart details for users. Primary key: cart_id

2. **Order Details** – Keeps track of products in each order. Attributes: id, quantity, product_id

3. **Products** – Stores information about farm products. Attributes: product_id, name, description, price, photo

4. **Review Table** – Stores user reviews and ratings for products. Attributes: id, datetime

5. **Sales** – Records sales transactions. Attributes: sales_id, payment_id, sales_date

6. **Users** – Stores user account information. Attributes: user_id, username, password, contact_details.

System Operations:

- **Admin:** Can add, update, or delete products and categories; monitor orders and sales; manage users.

- **Farmers/Users:** Can browse products, place orders, and provide reviews.

- **Marketplace:** Tracks stock levels, generates sales records, and maintains secure user and transaction data.

This ERD ensures that all necessary data for operations, orders, and transactions is structured efficiently, making the system scalable, secure, and easy to maintain.

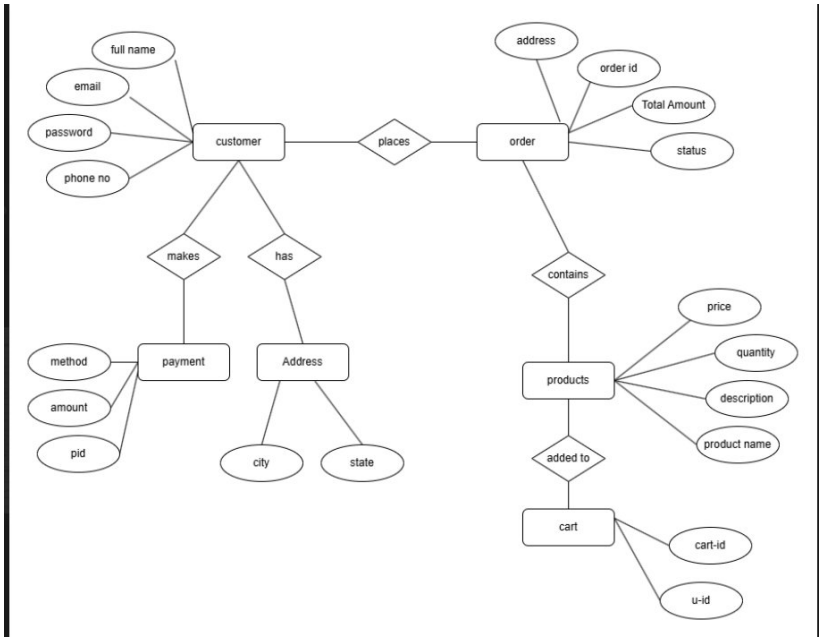

Figure 3 ER Diagram of Local Framer Marketplace

3.1.4 DFD (Data Flow Diagram)

Figure 4 DFD 0Levelof Local Framer Marketplace

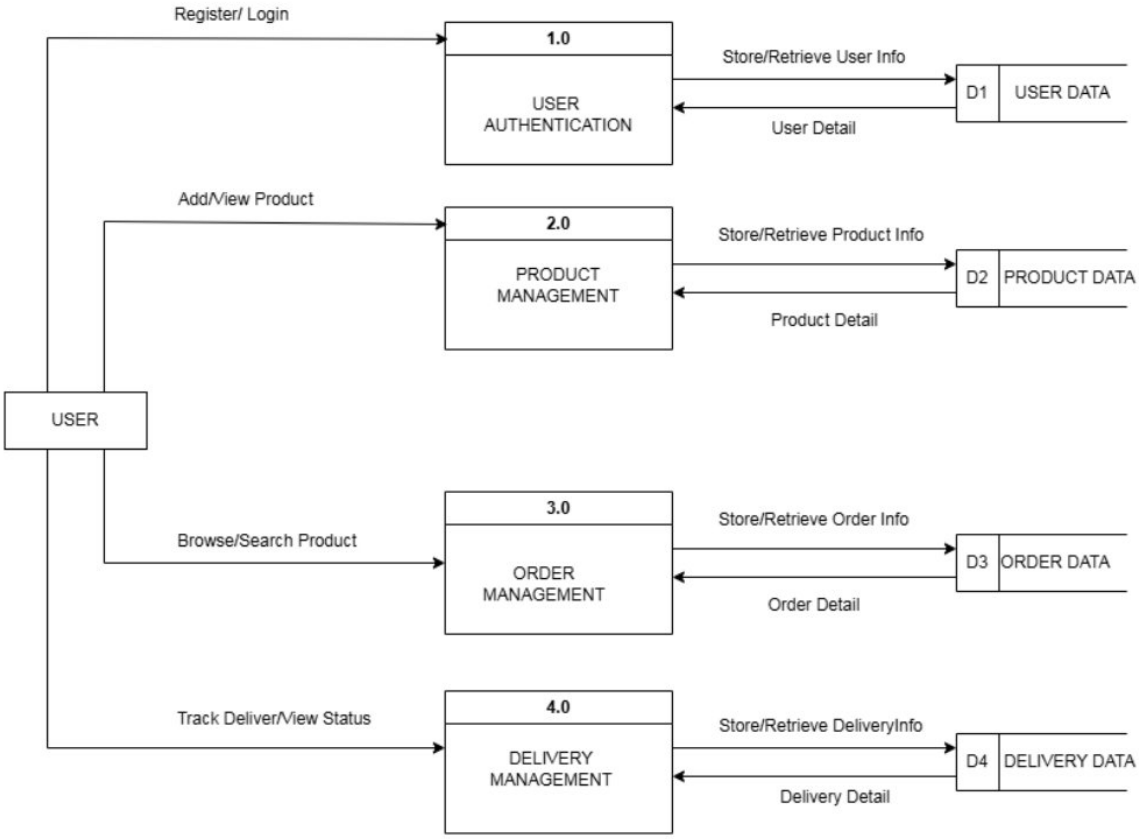

Figure 5 DFD Level1 of Local Framer Marketplace

3.2 System Design

The Local Farmer Marketplace System is designed as a web-based platform that connects farmers directly with customers, allowing easy buying and selling of agricultural products. The system architecture is structured into multiple layers to ensure efficiency, scalability, and smooth communication between users, farmers, and administrators. It follows a client-server architecture where users interact with the system through a web interface, while the server processes requests and manages data. The system includes a database layer that securely stores information such as user details, product listings, and transactions. An application layer handles business logic, ensuring proper functioning of features like product search, order management, and user authentication.

3.2.1 Methodology to be used

The proposed Local Farmers Marketplace system has been developed using the Agile model, an iterative and flexible software development approach that allows continuous refinement throughout the project. Development began with collecting and analyzing the needs of farmers and consumers such as product listing, order management, and buyer-seller communication and these requirements were implemented in small, manageable sprints. Each sprint included planning, designing, coding, testing, and reviewing, enabling regular feedback and improvements.

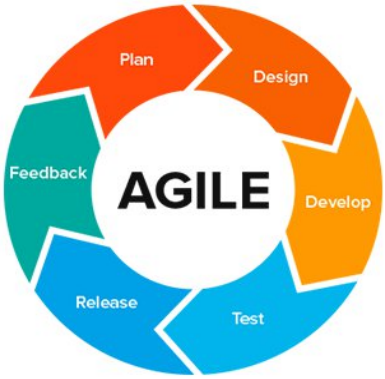

Figure 6 Agile Model

3.2.2 Architectural Design

We've separated our application into three tier. So, three-tier architecture has been used for this project. Client can't directly connect with the server, so we use three-tier technology which provides application layer section to connect client and the server.

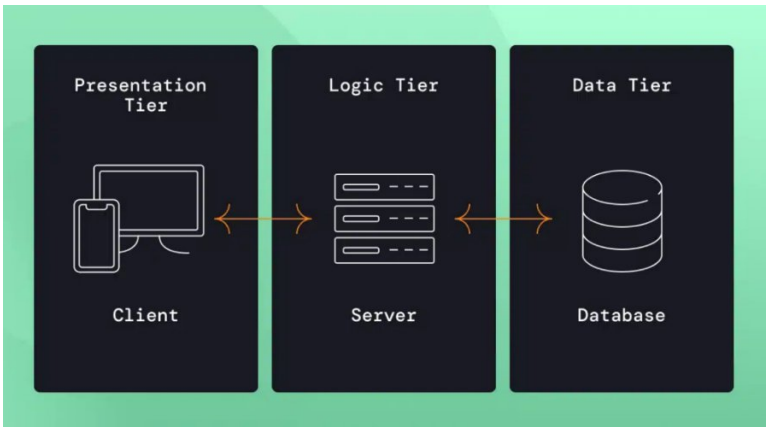

Figure 7 Three Tier Technology

Three-tier architecture has three layers:

- Presentation Tier- Comprises of the application GUI, shows information related to services inhabiting the top level. It interconnects with the other tier by sending request to the browser.

- Logic Tier- Characterizes a view of the database for a user. Its performances as the intermediary between the end-user and the database. It holds the business logic to regain data from the user requests.

- Database Tier- The data is stored and repossessed as it is the houses of the database servers.

Here the data is kept independent of the business logic or the application servers.

3-tier architecture contains an application layer between presentation layer and database server, which is responsible of making communication possible between client requests to the database and send the response result from database to the client.

The advantages of 3-tier architecture are:

- It supports Database Management System (DBMS)

- Program-data independence

- It supports multiple views of data

3.2.3 Flowchart

A flowchart is a visual representation that shows the sequence of processes, decisions, and actions within a system using standard symbols and arrows. This flowchart explains how users interact with the Local Farmers Marketplace. When a user visits the website, they first arrive at the homepage where they can browse available products. However, if they are not logged in, they may not be able to place orders or manage products. Users can also visit the About page to learn more about the platform and its purpose. If a user wants to buy or sell products, they must log in by entering their email and password. If the credentials are incorrect, access is denied; if correct, they are redirected to their dashboard. New users who do not have an account can register by providing necessary details such as name, email, password, and role (farmer or customer).Once logged in, farmers can add, update, or delete product listings, while customers can browse products, place orders, and view their order history. Users can also update their profile information or change their password. The process ends when the user logs out or exits the website.

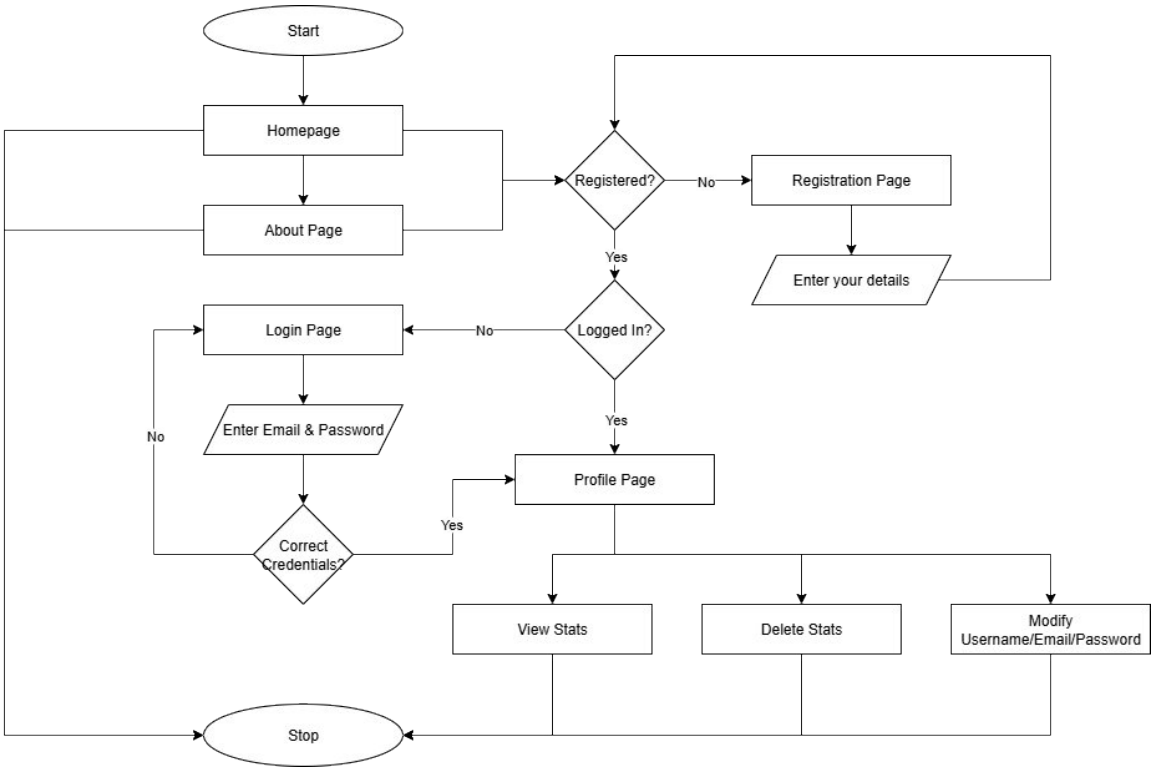

Figure 8 System Flow Chart

3.3 Algorithms Used in Local Farmer Marketplace

The system uses different algorithms to improve product search, sorting, and inventory management. These algorithms help make the marketplace faster, more organized, and easier for both users and farmers.

1. Product Search Algorithm

The system uses the Linear Search Algorithm to help users find products easily. It checks each product one by one in the product list until the required item is found.

In the local farmer marketplace, this algorithm is used to search agricultural products such as vegetables, fruits, grains, and other farm goods based on keywords entered by the user. It compares product names and descriptions sequentially to return relevant results.

2. Product Sorting Algorithm

The system uses the QuickSort Algorithm to arrange products in an organized way. It works by selecting a pivot element and dividing the list into smaller and larger values, then sorting them recursively.

In this marketplace, QuickSort is used to sort products by price, name, or availability. This allows users to easily compare farm products and choose items based on their preferences, making the shopping experience more efficient.

3. Stock Management Alert Algorithm

The system includes an Inventory Alert Algorithm to monitor product stock levels. It automatically checks whether the stock of any product falls below a defined minimum level.

In the local farmer marketplace, this algorithm alerts farmers or administrators when agricultural products such as vegetables or fruits are running low. This helps ensure timely restocking and prevents product shortages in the marketplace.

CHAPTER 4:

IMPLEMENTATION AND TESTING

4.1 Implementation

4.1.1 Tools Used (CASE tools, Programming language, Database platforms)

The following tools and frameworks were used for the development of the Local Farmer Marketplace system:

Front-End Tools

- **HTML**

In the Local Farmer Marketplace, HTML is used to create and structure web pages. It defines the layout of the system such as product pages, login pages, and dashboards. HTML elements like `<form>`, `<input>`, `<button>`, and `<select>` are used to collect user information and display farm products such as vegetables, fruits, and grains.

- **CSS**

In the Local Farmer Marketplace, CSS is used to design and style the web pages. It improves the appearance of the system by controlling colors, fonts, layouts, and spacing. CSS ensures that the marketplace interface is attractive, user-friendly, and responsive for both farmers and buyers.

- **JavaScript**

In the Local Farmer Marketplace, JavaScript is used to make the website interactive and dynamic. It is used for form validation, real-time updates, and improving user experience. JavaScript helps in making features like product search, cart updates, and alerts more responsive.

Back-End Tools

- **PHP**

In the Local Farmer Marketplace, PHP is used for server-side development. It handles database connectivity, user authentication, and data processing. PHP is responsible for managing user registration, login, product management, order processing, and performing CRUD operations (Create, Read, Update, Delete) on marketplace data.

Server

- **Apache Server**

In the Local Farmer Marketplace, Apache Server is used to run PHP scripts and handle server-side operations. It processes user requests and ensures smooth communication between the front-end and the database, enabling dynamic web page generation.

Database

- **MySQL**

In the Local Farmer Marketplace, MySQL is used to store all system data such as user details, farmer information, product listings, orders, and transactions. It supports CRUD operations, ensuring efficient data management and retrieval for the marketplace system.

Documentation Tools

- **MS Office**

MS Office is used for preparing reports, documentation, and maintaining project records and progress tracking during system development.

- **Draw.io**

Draw.io is used to create system diagrams such as DFDs, ER diagrams, and flowcharts for the Local Farmer Marketplace. It helps in designing system architecture easily using drag-and-drop features.

4.1.2 Implementation Details of Modules

The Local Farmer Marketplace System consists of multiple modules, each designed to support smooth interaction between farmers, buyers, and the system administrator. The main modules are described below:

1. Admin Module

- **Product Management:**

The admin can add, update, and delete agricultural products such as vegetables, fruits, grains, and dairy items. Each product includes details like name, price, quantity, category, description, and images.

- **User & Farmer Management:**

The admin can manage registered users and farmers by verifying accounts and maintaining system records.

- **Order Management:**

The admin can monitor and manage all customer orders, including updating order status such as pending, confirmed, packed, and delivered.

2. User (Buyer) Module

- **User Registration & Login:**

Buyers must register by providing details such as name, email, password, and contact information. After successful registration, they can log in to access the system.

- **Product Browsing:**

Users can browse available farm products based on categories like vegetables, fruits, grains, and organic products.

- **Shopping Cart:**

Users can add selected agricultural products to their cart and manage quantities before placing an order.

- **Order Placement:**

Users can confirm their cart items and place orders. The system forwards the order to farmers/admin for processing and delivery.

3. Farmer Module

- **Farmer Registration & Login:**

Farmers register by providing farm details, contact information, and product categories. After approval, they can log in to the system.

- **Product Management:**

Farmers can add their farm products, update stock, change prices, and remove unavailable items.

- **Order Handling:**

Farmers can view incoming orders, prepare products, and update order status for delivery.

4. Login Module

This module allows both users (buyers), farmers, and admins to securely log into the system using valid credentials. The system verifies login information to ensure authorized access and protect user data.

5. Registration Module

New users and farmers must register by providing necessary details such as name, email, password, address, and contact number. After successful registration, users can access system features like browsing products, placing orders, and managing their profiles.

4.1.3 Test Cases for Unit Testing

In system testing, individual modules of the Local Farmer Marketplace System are tested to ensure they work correctly as shown below:

Table 1 *Test Case for Registration of User*

S.N	Test Name	Input	Expected Output	Actual Output	Test Result
1	Open Application	http://localhost/Local-Farmer-Marketplace	Registration Page	Registration Page	Pass
2	Enter Invalid Details	Email: test123, Name: user, Password: 12345	Invalid Email	Invalid Email	Pass
3	Enter Valid Details	Email: user@gmail.com, Name: user, Password: password	Registration Successful	Registration Successful	Pass

Table 2 *Test Case for Login*

S.N	Test Name	Input	Expected Outcome	Actual Output	Test Result
1	Open Application	http://localhost/Local-Farmer-Marketplace	Login Page	Login Page	Pass
2	Enter Invalid Credentials	Email: user@gmail.com, Password: 12345	Incorrect Password	Incorrect Password	Pass
3	Enter Valid Credentials	Email: user@gmail.com, Password: password	Login Successful	Login Successful	Pass

Table 3 Test Case for Adding Product (Farmer/Admin)

S.N	Test Name	Input	Expected Outcome	Actual Output	Test Result
1	Open Product Page	http://localhost/Local-Farmer-Marketplace/add_product.php	Product Page	Product Page	Pass
2	Invalid Product Entry	Name: Tomato, Price: blank, Category: Vegetable	Please fill out this field	Please fill out this field	Pass
3	Valid Product Entry	Name: Tomato, Price: 80, Category: Vegetable, Image: tomato.png	Product added successfully	Product added successfully	Pass

Table 4 Test Case for User Actions

Test Case	Expected Data	Test Result
Click Home	View homepage	Successful
Click Products	View all farm products	Successful
Click Product Details	View selected product information	Successful
Click About Us	View system information	Successful
Click Blog	View agriculture-related blog section	Successful
Click Orders	View user order history	Successful

Table 5 ***Test Case for Admin Actions***

Test Case	Expected Data	Test Result
Click Home	View admin dashboard	Successful
Click Products	Add, update, delete farm products	Successful
Click Orders	Manage customer orders	Successful
Click Users/Farmers	View and manage registered users and farmers	Successful

CHAPTER 5:

CONCLUSION AND FUTURE RECOMMENDATIONS

5.1 Lesson Learnt / Outcome

Every project provides valuable learning experiences and helps improve technical as well as soft skills. Through the development of the Local Farmer Marketplace System, we gained knowledge in system development, teamwork, and practical problem-solving. The key lessons learned are described below:

• Team Collaboration

This project was developed as a group effort, which helped us understand the importance of coordination and cooperation. We learned how to divide tasks among team members such as design, coding, testing, and documentation. Working together also improved our ability to discuss problems and find solutions collectively during system development.

• Problem-Solving Ability

During the development of the marketplace system, we faced various technical issues such as database errors, login issues, and UI design problems. This helped us improve our analytical thinking and debugging skills. We learned how to identify errors step by step and resolve them efficiently.

• Documentation and Technical Writing

We gained experience in preparing project reports, system documentation, and diagrams such as ER diagrams, DFDs, and system flowcharts. This improved our ability to present technical information in a structured and clear manner using proper formatting and CASE tools.

• Time Management

One of the most important outcomes of this project was learning how to manage time effectively. We understood how to prioritize tasks such as backend development, frontend design, and testing to complete the system within the given deadline without affecting quality.

• System Development Knowledge

Through this project, we also gained practical knowledge of building a real-world application, including database handling, user interaction, and system integration. It helped us understand how an online marketplace for farmers operates in a digital environment.

5.2 Conclusion

The Local Farmer Marketplace System has been successfully developed as a web-based platform that connects local farmers directly with customers. The system fulfills the main objectives defined in the planning phase, including secure user and farmer authentication, product management, categorized agricultural listings, search functionality, order placement, and inventory tracking.

The platform provides a simple and user-friendly interface where customers can easily browse farm products such as vegetables, fruits, grains, and dairy items. Users can view product details, compare prices, and place orders conveniently without any complexity.

On the other hand, farmers and administrators can efficiently manage product listings, update stock availability, and handle incoming orders through the admin panel. This ensures smooth coordination between supply and demand within the system.

Overall, the Local Farmer Marketplace successfully demonstrates how digital technology can support agriculture by improving accessibility, reducing middlemen, and creating a direct connection between farmers and consumers. It provides a practical, efficient, and scalable solution for modern agricultural product distribution.

5.3 Future Recommendations

Although the Local Farmer Marketplace System is fully functional and achieves its main objectives, there are several improvements that can be implemented in future versions to enhance usability, performance, and overall efficiency.

1. Multiple Payment Integration

In future development, the system can support additional payment methods such as Khalti, mobile banking, and other digital wallets. This will make transactions more flexible and convenient for users and farmers.

2. Real-Time Order Tracking

An advanced tracking system can be added where users can monitor their orders from confirmation to delivery. Status updates like *Confirmed, Packed, Dispatched, and Delivered* will improve transparency and trust in the system.

3. Improved Security Features

To make the platform more secure, features such as OTP verification, two-factor authentication (2FA), and stronger password encryption can be implemented. This will help protect user accounts and sensitive data.

4. Smart Product Suggestions

A recommendation system can be introduced to suggest products based on user search history and previous purchases. This will improve user experience and help farmers sell more products.

5. Mobile Application Development

Developing mobile applications for Android and iOS will make the Local Farmer Marketplace more accessible. It will allow users and farmers to manage activities easily from their mobile devices.

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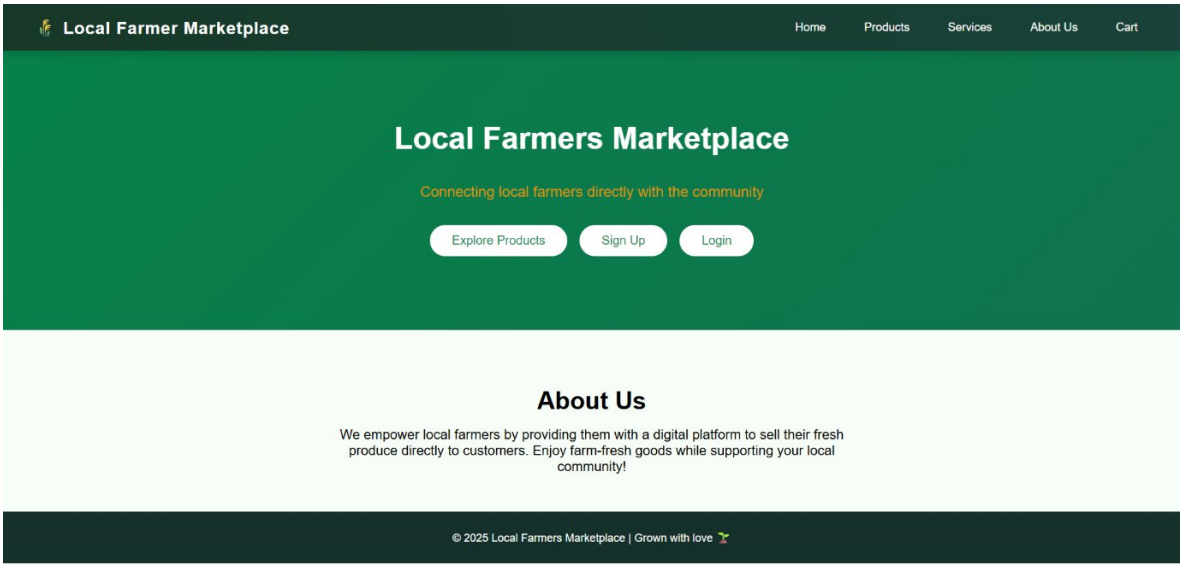
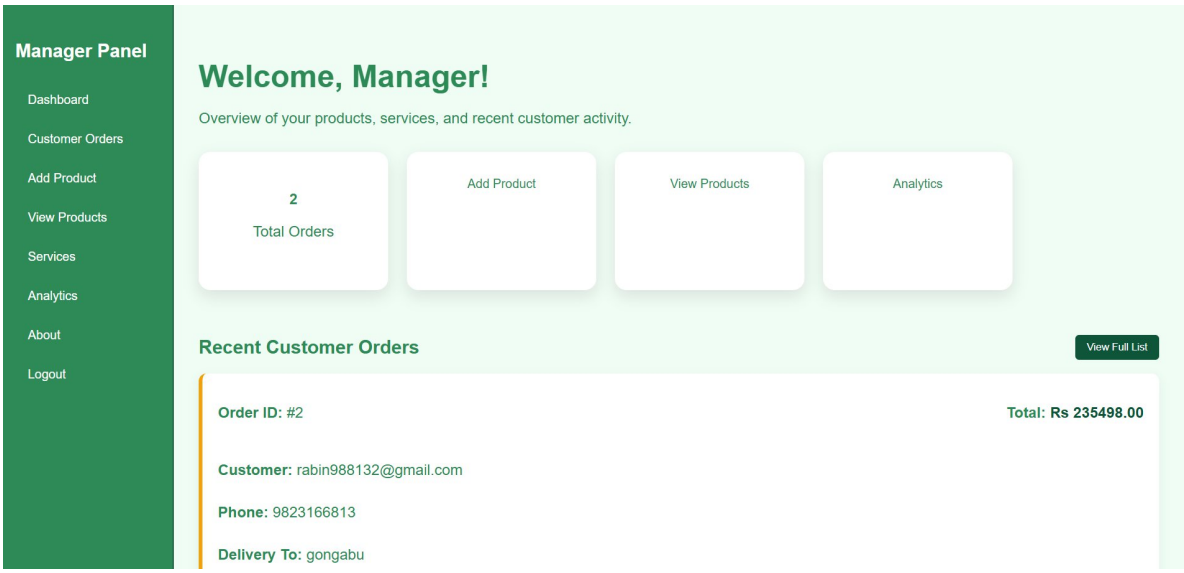
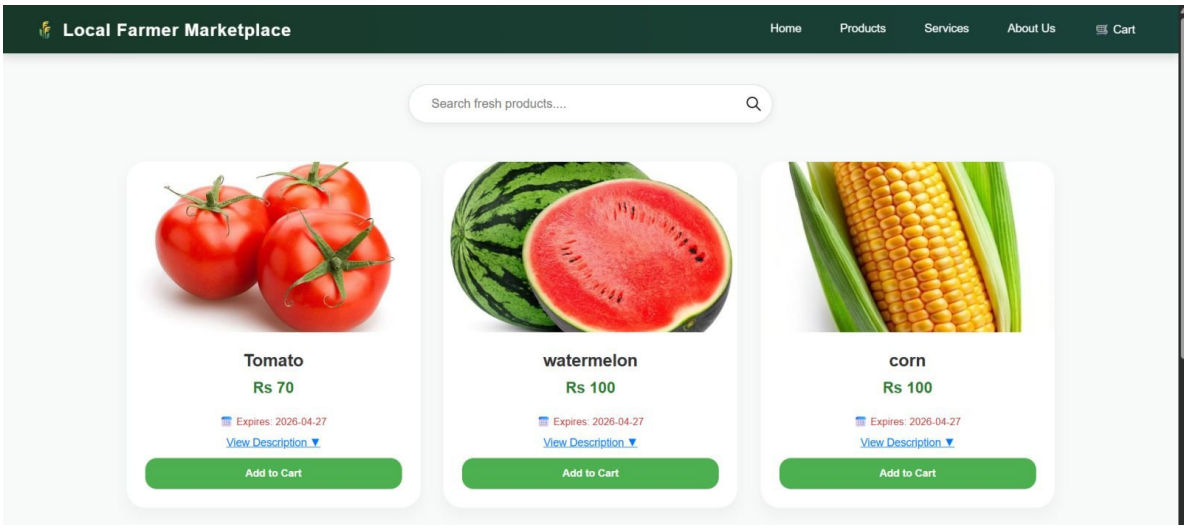
APPENDIX: SYSTEM SCREENSHOT






Local Farmer Marketplace
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Products
Services
About Us
Cart
Search fresh products....
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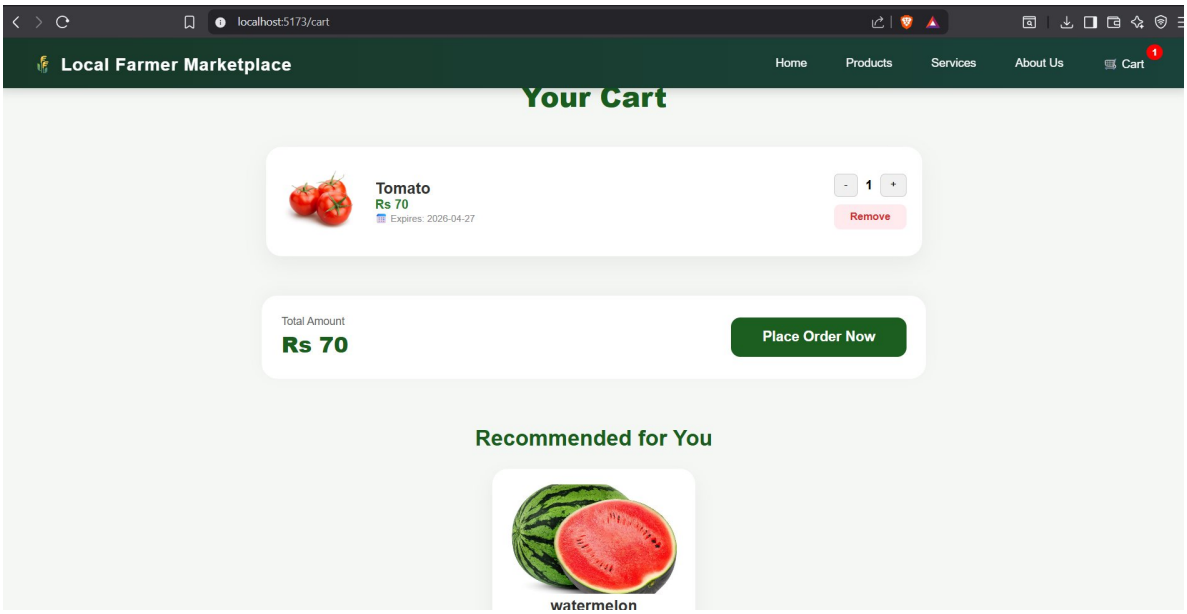
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Services
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Total Amount
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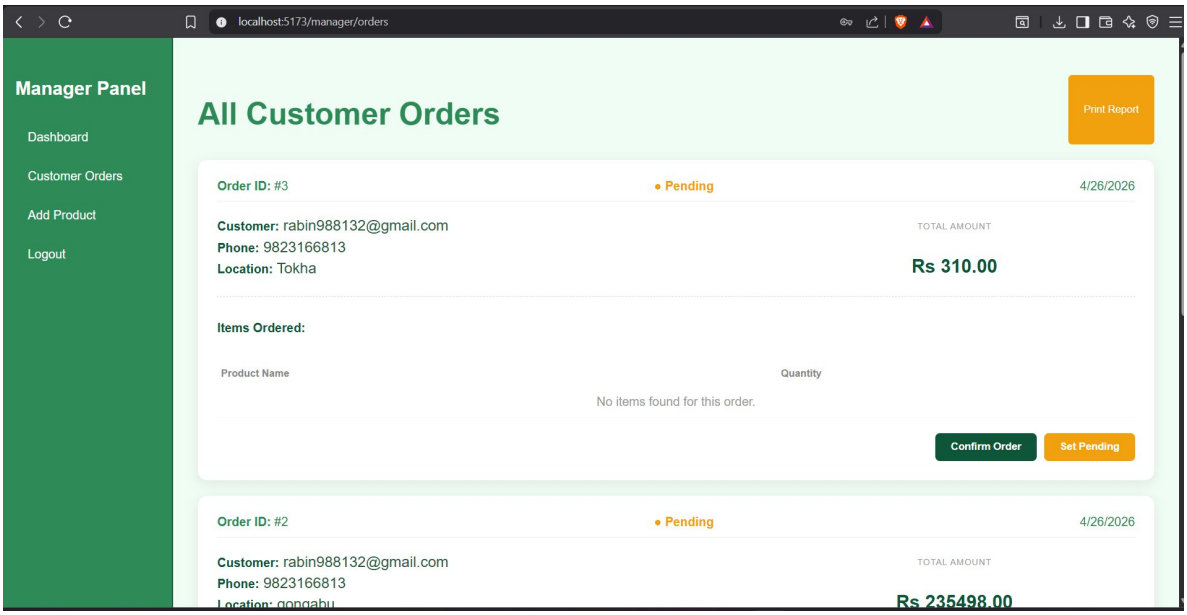
Place Order Now
Recommended for You
watermelon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Customer Orders
Add Product
Logout
All Customer Orders
Print Report
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• Pending
4/26/2026
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Phone: 9823166813
Location: Tokha
TOTAL AMOUNT
Rs 310.00
Items Ordered:
Product Name
Quantity
No items found for this order.
Confirm Order
Set Pending
Order ID: #2
• Pending
4/26/2026
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Phone: 9823166813
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